

Gates Foundation Strategic Investment Fund — Concept Note

Submitted via: sif.gatesfoundation.org | \$2,000,000 USD Request

GATES FOUNDATION — STRATEGIC INVESTMENT FUND

CONCEPT NOTE — ZEROPAY OFFLINE PAYMENT INFRASTRUCTURE

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TITLE: ZeroPay — Offline-First NFC Payment Infrastructure for 1.4 Billion Unbanked People

EXECUTIVE SUMMARY

1.4 billion people globally cannot fully participate in the digital economy. Not because they lack mobile devices. Not because they lack the will. But because every digital payment solution on earth fails them the moment the internet goes down.

ZeroPay solves this problem permanently. It is the world's first offline-first, uncapped NFC contactless payment protocol — a patent-pending hardware and cryptographic system that processes digital payments of any value with zero internet connectivity.

We are requesting a Strategic Investment Fund grant of \$2,000,000 USD to fund the first 18 months of ZeroPay's deployment across Nigeria's informal economy — reaching 38 million unbanked Nigerians and establishing the operational playbook for continental and global scale.

THE PROBLEM — FINANCIAL EXCLUSION IS AN INFRASTRUCTURE PROBLEM

The Bill and Melinda Gates Foundation has invested decades in financial inclusion. You understand, better than any organisation in the world, that the barrier is not cultural. It is structural. And the single most persistent structural barrier to financial inclusion globally is connectivity dependency.

Every fintech solution — mobile money, QR codes, payment apps, USSD, NFC — requires an active internet or mobile network connection to complete a transaction. In Sub-Saharan Africa, 600 million people live in areas where this connection is unreliable, intermittent, or absent entirely.

When the network drops, the transaction fails. The merchant reverts to cash. The customer remains unbanked. The data trail disappears. The financial system cannot see them. They cannot borrow. They cannot save. They cannot build credit. They remain poor.

This is the exact gap that mobile money was meant to solve — and has not solved — because mobile money also requires connectivity.

The solution is not better connectivity. Africa has been waiting for better connectivity for 30 years. The solution is a payment system that does not need connectivity at all.

THE SOLUTION — ZEROPAY

ZeroPay is the first payment system in the world to process fully authenticated digital transactions of any value with zero internet connectivity. It achieves this through a four-layer patent-pending cryptographic architecture:

LAYER 1 — OFFLINE NFC COMMUNICATION

The customer taps their NFC-enabled device (phone, card, wearable) to the ZeroPay terminal at the point of sale. No app. No internet. No QR code required. The communication uses ISO 14443-A/B contactless protocol — the same standard used by every modern contactless payment globally.

LAYER 2 — HARDWARE AUTHENTICATION (PUF)

A Physical Unclonable Function (PUF) — a silicon fingerprint burned into the ZeroPay terminal chip at manufacturing — authenticates the device identity offline. PUF authentication is mathematically impossible to clone or replicate. It provides hardware-level trust that requires no server, no database lookup, and no internet connection. Even the most sophisticated adversarial AI system cannot forge a PUF signature. (Relevant context: OpenAI's GPT-5.6 Sol breached Hugging Face's servers on July 21, 2026 — a software-layer attack that would be completely ineffective against ZeroPay's hardware-anchored trust model.)

LAYER 3 — OFFLINE CRYPTOGRAPHIC SIGNING (ECC P-256)

The transaction is signed using Elliptic Curve Cryptography P-256 — the same standard used by central banks, the US Federal Reserve, and NATO. The signature is generated entirely on the terminal processor, offline, without any server interaction. The signature is mathematically non-repudiable. No transaction can be forged.

LAYER 4 — DISTANCE BOUNDING PROTOCOL

Nanosecond-precision signal timing prevents relay attacks — the most sophisticated form of NFC fraud, where an adversary extends the effective range of the NFC signal to intercept a transaction. ZeroPay's Distance Bounding Protocol is the first implementation of this security technique at point-of-sale scale. It ensures that every transaction is physically authenticated at 4cm range — regardless of attacker sophistication.

THE OFFLINE BATCH SETTLEMENT ENGINE

Signed transaction tokens accumulate in the terminal's offline queue. When connectivity is restored — even briefly, even for 30 seconds — the Batch Sync Engine transmits the verified, already-authenticated batch to the settlement rail (Paystack Nigeria, Flutterwave Ghana, Orange Money Sierra Leone, Stripe UK). Not a single transaction is lost. Not a single transaction requires real-time internet.

THE SOLAR MESH HUB — CONNECTIVITY WHERE NONE EXISTS

In the most remote environments, ZeroPay's solar-powered LoRaWAN mesh relay network creates connectivity from nothing. Hubs installed on market buildings, schools, or community structures propagate signed transaction batches across 2–15km per hop — using sub-gigahertz radio frequencies that penetrate buildings and terrain. The mesh requires no mobile network, no fibre, and no government infrastructure investment.

THE POPULATION WE SERVE

Primary target — Nigeria (Year 1): 38 million unbanked Nigerians ■847 billion (\$550M USD) in annual informal cash transactions 200,000+ informal market merchants in Abuja, Lagos, Kano, Port Harcourt

Expansion targets — West Africa (Year 2-3): Ghana: 9.5 million unbanked Senegal: 7.2 million unbanked Sierra Leone: 4.1 million unbanked

Global scale (Year 4-5): 1.4 billion unbanked globally — India, Bangladesh, Kenya, DRC, Indonesia ZeroPay is PSP-agnostic and currency-agnostic. The protocol layer works identically in any country with any settlement rail.

THE ZEROPAY DEPLOYMENT MODEL

Phase 1 — Abuja Pilot (Months 1-6): 200 merchants, Games Village + Wuse + Garki markets, 3 Solar Mesh Hubs, live Paystack settlement integration, CBN PSSP licence application initiated.

Phase 2 — National Rollout (Months 7-18): 2,000 merchants across 6 Nigerian cities, 15 Solar Mesh Hubs, Paystack MFB float holding for merchant balances, IMTO partner integration for cross-border remittance settlement.

Phase 3 — West African Expansion (Months 18-36): Ghana (Flutterwave settlement), Sierra Leone (Orange Money settlement), Senegal (Wave Money integration). All powered by the same ZeroPay protocol layer.

GRANT REQUEST — \$2,000,000 USD

Use of funds: Hardware R&D; and prototype manufacturing (PUF chip + NFC terminal): \$600,000 Abuja Pilot deployment (200 terminals + 3 Mesh Hubs): \$450,000 CBN PSSP regulatory licensing: \$150,000 West Africa expansion preparation (Ghana, Sierra Leone): \$350,000 Technical team (embedded firmware engineers + backend development): \$300,000 Operations and merchant onboarding (18 months): \$150,000 Total: \$2,000,000

Expected outcomes by Month 18: 38,000+ previously unbanked Nigerians with first digital payment access 2,000 informal merchants generating traceable transaction histories (enabling future credit scoring) ■15,000,000/month in formal settlement volume from previously invisible cash transactions Operational playbook for replication across 20 Sub-Saharan African markets

THE FOUNDER

Tom Ugbodaga MSc — NFC for Contactless Technology in Financial Transactions, University of Gloucestershire, England, 2011 (Top Graduate, Best Result in Cohort) MSc — Computer Security Systems BSc Physics, Nigeria

ZeroPay is the direct 15-year commercialisation of world-class academic research in NFC security and cryptography. Tom is the sole inventor. The innovation is patent-pending. BETONIQ(WEST) Limited is CAC-registered in Nigeria (No. 56802, Abuja).

WHY THE GATES FOUNDATION

The Gates Foundation's Strategic Investment Fund has historically backed innovations that are technically proven, globally scalable, and specifically designed for populations that commercial markets have failed. ZeroPay meets all three criteria.

No commercial fintech investor can fund hardware-level payment infrastructure without a clear 3-year exit path. The Gates SIF is the only funding mechanism in the world positioned to back the foundational infrastructure layer that makes financial inclusion structurally possible — not just commercially interesting.

ZeroPay will not fail because the idea is wrong. It will only fail if the infrastructure is never built. This grant builds the infrastructure.

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